

Troubleshooting Log

Secure Enterprise Network System. Problems encountered during the build, how each was isolated, and what fixed it. Cases are listed in the order they came up.

Case 1: IP phones assigned the wrong kind of VLAN

Symptom. Phones connected to access ports were not registering or handling voice traffic correctly.

Root cause. The phone-facing ports were configured with `switchport access vlan 70`, which treats the voice VLAN as a plain data VLAN. Phones expect a voice VLAN, which rides on the port alongside the data VLAN and lets the phone tag its own traffic.

```
interface range fa0/5-6
  no switchport access vlan 70
  switchport voice vlan 70
```

Takeaway. Access VLAN and voice VLAN are different features. A phone port normally carries the PC's data VLAN untagged plus the voice VLAN tagged on the same interface.

Case 2: Dead link toward ISP 1

Symptom. No connectivity across the edge link to the first ISP.

Root cause. Two problems stacked: the router interface was administratively shut down, and the subnet mask on the point-to-point link did not match the far side (it needed to be /30).

Fix. Brought the interface up with `no shutdown` and corrected the mask on both ends of the link.

Takeaway. Check interface state first, then confirm both ends of a link agree on the subnet. A mismatched mask can break a link that shows up/up.

Case 3: Clients could not reach DHCP

Symptom. PCs failed to obtain addresses even though the DHCP server was running.

Root cause. DHCP discovery is a broadcast, and broadcasts do not cross a Layer 3 boundary. The gateway SVIs had no relay configured, so requests never reached the server's subnet.

```
interface vlan 20
  ip helper-address 10.11.11.38 ! DHCP server
```

Fix. Added the helper address on every client-facing SVI on both core switches, which forwards DHCP broadcasts to the server as unicast.

Takeaway. Whenever the DHCP server sits in a different subnet than the client, a relay on the gateway is mandatory.

Case 4: Some subnets still received no leases (end-to-end isolation)

Symptom. After the relay fix, LAN and WLAN clients still received no addresses. This turned into a full layer-by-layer isolation exercise:

- **Access switch:** confirmed the client port was connected (`show interfaces status`), in the right VLAN (`show vlan brief`), and that the VLAN was allowed on the trunk (`show interfaces trunk`).
- **Core switch:** verified the SVIs were up (`show ip interface brief`), HSRP was healthy (`show standby brief`), the correct helper address was set, and the core could ping the DHCP server.

- **Firewall:** confirmed interfaces were up and the routing table covered all internal networks.
- **Server side:** confirmed the server could reach its own gateway and the client VLAN gateway, proving end-to-end connectivity was fine.
- **Server configuration:** listed the DHCP pools and found only the Management and Server pools existed. No pools had been created for the LAN or WLAN client subnets at all.

Root cause. The DHCP server had no scopes for those subnets, so it had nothing to offer.

Fix. Created the missing LAN and WLAN pools.

Takeaway. Prove each hop one layer at a time, and never assume a service is fully configured just because the server answers ping.

Case 5: OSPF neighbors not forming

Symptom. Expected routes were missing and the adjacency never established between two devices.

Root cause. An OSPF configuration mismatch between the neighbors.

Fix. Aligned the OSPF configuration on both sides and confirmed the adjacency reached FULL with `show ip ospf neighbor`.

Takeaway. Directly connected networks never appear in `show ip route ospf`; they show as connected routes in the full table. Their absence from the OSPF-only view is not a failure.

Case 6: First ping fails, the rest succeed

Symptom. The first ICMP echo to a new destination timed out, and every ping after that succeeded.

Cause. Layer 3 devices along the path must resolve the next hop's MAC address with ARP before forwarding. The first packet can expire while that resolution completes. Once the ARP tables are populated, forwarding is immediate.

Takeaway. Expected behavior, not a fault.

Case 7: Access point advertising no WLANs

Symptom. An AP had power and link but broadcast no SSIDs.

Root cause. The AP's switch port was assigned Access VLAN 1 with Voice VLAN 70 instead of the VLAN the AP actually needed.

Fix. Corrected the port's VLAN assignment; the AP immediately displayed the configured WLANs and wireless connectivity was restored.

Case 8: Lightweight AP never registering with the WLC (CAPWAP)

Working through a stuck AP registration produced this checklist, in order:

- WLC management interface has the correct IP, mask, gateway, and VLAN.
- WLC management IP answers ping from a PC or the core switch.
- The WLAN (SSID) is actually enabled on the WLC.
- The DHCP pool hands out the WLC address (Option 43) so APs can locate the controller.
- The AP's switch port is connected, enabled, and in the correct management VLAN.
- The management VLAN SVI is up and carries the right `ip helper` -address when DHCP lives in another subnet.

- The DHCP server is issuing leases for the AP's VLAN.
- The AP obtains an IP and its CAPWAP state changes to Connected.
- The AP appears on the WLC with a real IP instead of 0.0.0.0.
- If everything above checks out and it still fails in Packet Tracer, remove and re-add the AP; the simulator occasionally leaves APs stuck in a failed CAPWAP state.

Design points confirmed while troubleshooting

- **HSRP addressing.** Each VLAN ends up with three addresses: one shared virtual gateway (the first address, used by every host) and one real SVI address per core switch. As a convention, the standby switch took the second address and the active took the third. HSRP does not require this, but a consistent scheme makes the intended active switch obvious at a glance. Priority decides the active role and preempt lets the preferred switch reclaim it after recovery.
- **Why servers never use DHCP.** DMZ servers (Web, Mail, FTP, App, NAS) keep static addresses so NAT rules and firewall policy always point at the right host. Internal servers (DHCP, DNS, RADIUS) keep static addresses so clients and network devices can always find them. DHCP stays a client-only mechanism, since client addresses can change without breaking anything.
- **Management scope.** SSH was deployed on the switching infrastructure and locked to the management subnet with a VTY access-class. In a production network, routers and firewalls receive the same treatment.

The method

Every problem above fell to the same approach: start at the access port, prove the Layer 2 path, prove the gateway, prove routing, prove the service host, and only then question the application configuration. Isolating one hop at a time shrinks the search space with every verified step.